

# Governed Agent Morphogenesis: Runtime Organizational Reconfiguration with Bounded Authority and Causal Continuity

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Reference implementation: AgentPlat Agent Morphogenesis.

Status: systems protocol with reproducible local boundary, integration and finite-grid evaluations.

## Abstract

Agent collectives may need to recruit specialists, transfer work, and retire participants during a mission. Such changes cross independently authoritative services and can leave material effects even when a competing proposal wins organizational commitment. We present a transition contract connecting an exact approved proposal, current owner authority, stable effect identities, accepted lineage, and explicit cleanup. A reference implementation separates successful completion from terminal supersession of a losing proposal. Reproducible PostgreSQL experiments exercise concurrent transitions, mandate expiry, lost acknowledgements, and retirement with admitted work. An adversarial admission matrix compares four controllers under identical owner enforcement; the durable workflow and Morphogenesis both reject invalid authority and preserve recoverable effects. A bounded TLA+ model checks successor uniqueness, stable effects and safe termination, with counterexamples for deliberately weakened variants. These results identify the integration obligations that make organizational reconfiguration inspectable and recoverable.

**Keywords:** multi-agent systems; organizational adaptation; agent governance; durable execution; causal continuity; runtime reconfiguration.

## 1. Introduction

Consider a person and an agent collective preparing a technical proposal. Its initial members can research requirements and draft an answer, but a new requirement needs a specialist. The collective may recruit an existing agent or instantiate a temporary one, assign bounded work, incorporate its output, and eventually remove it. Even this small adaptation crosses several boundaries: a factory materializes an agent, membership admits an identity, a team incorporates a participant, work contracts allocate responsibility, and an action gateway determines which external effects are permitted.

Changing a list of agents does not coordinate those boundaries. A decision may expire before it is used. Two controllers may propose different successors from the same organization. A provider may perform an effect and lose its acknowledgement. Retirement may remove a resource while leaving actionable work behind. A resumed controller may encounter evidence that an operation was prepared without knowing whether it completed.

This paper asks: **under which explicit assumptions can an agent collective transform its operational organization while preserving bounded authority and a reconstructible account of ongoing work?** We use *morphogenesis* to denote mission-driven changes in operational form. The term is an engineering abstraction; it does not imply biological development, autonomous evolution of model weights, or an intrinsic tendency toward better organizations.

Organizational modeling, architectural adaptation, and compensation already provide important foundations. AgentPlat's proposed contribution is a concrete composition: each organizational transition connects observation, intent, authorization, owner-specific effects, recovery, and a versioned morphology head. The morphology controller coordinates these owners without replacing their authority.

The paper makes three contributions: (1) a transition model separating organizational intent, decision authority, and effect ownership; (2) a recoverable protocol with explicit assumptions and implementation obligations for successor selection, retirement, and evidence continuity; and (3) an open-source reference mapping and reproducible boundary, integration and four-condition mission evaluations, including a counterexample to

interpreting head consistency as atomic reconfiguration. Protocol safety and organizational utility are evaluated as separate questions.

The scope is deliberately bounded. The core protocol concerns recruitment, catalog-based instantiation, detachment, and retirement, with selected advanced operators illustrating composition. Strategy learning, profile synthesis, whole-collective evolution, and constitutional amendment are extensions rather than prerequisites for the claims made here. No new model training algorithm or universal organizational optimizer is proposed.

## 2. Related work and positioning

### 2.1 Explicit agent organizations

Agent organizations have long been modeled through roles, groups, missions, permissions, and obligations. Moise separates structural, functional and deontic organizational specifications [13]. Dignum and Dignum relate organizational structure, objectives, and agent actions in a logical framework. These precedents mean that explicit organization and normative constraints are established research topics, not novelty claims of this paper. [2,13].

Runtime reorganization itself is also established. Sorici et al. describe a JaCaMo ReorgBoard, an implementation plan, and organizational rules for coordinating reorganization. Their plan separates stopping an organizational entity, changing its specification, and starting its successor [10, Sections 3-4]. ParaMoise represents reorganization through workflows and read/write locks over the affected parts of an organization; compatible reorganizations can proceed concurrently [11, Section 3.3]. These are direct predecessors, not merely background on static roles. [10,11].

Our focus is a particular cross-service contract: bind an exact organizational decision to owner-admitted effects, preserve their identities across uncertain outcomes, and record accepted lineage without treating that lineage as a global transaction. The comparison below identifies mechanisms described in the consulted sources. It does not claim their implementations lack unreported controls.

| Precedent                  | Mechanism established in the source                                | Boundary addressed here                                                                   |
|----------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| JaCaMo reorganization [10] | Norm-regulated reorganization and coordinated implementation plans | Stable effect identity and ambiguous outcomes across independently authoritative services |
| ParaMoise [11]             | Workflow-based reorganization and scoped read/write exclusion      | Receipt-linked successor commitment when owner effects are not globally atomic            |
| Sagas [8]                  | Compensating partially completed long-lived transactions           | Organizational admission, effect authority and post-commit lineage                        |
| MorphAgent [5]             | Feedback-driven profile adaptation                                 | How a selected change is authorized and recovered                                         |

We do not claim to originate organizational adaptation, normative governance, durable execution, or artifact-oriented collaboration.

### 2.2 Adaptation in LLM-based collectives

AgentVerse dynamically adjusts group composition through its collaboration process. MorphAgent evolves profiles and roles using task feedback. G-Designer constructs task-conditioned communication topologies. These works motivate adaptive collectives and provide relevant comparisons for future utility experiments. [4-6].

The detailed profile-update mechanism in MorphAgent v2 (Section 3) informs the comparison above; a profile proposal does not itself specify how heterogeneous effect owners recover an interrupted change. This is a scope distinction, not evidence that another implementation lacks recovery controls. Adaptation algorithms could supply candidate organizations to Morphogenesis, provided their proposals remain subject to the same admission rules. The mission study holds the adaptive proposal schedule constant. Cross-framework benchmarking would introduce additional integration confounders and is not performed here.

## 2.3 Architectural adaptation and durable effects

Rainbow uses architectural models and external adaptation mechanisms to support self-adaptive software. Morphogenesis similarly makes the managed structure explicit, while specializing the transition boundary to agent membership, work, and action authority. The adaptation-control pattern is inherited; the proposed contribution lies in how its agent-specific responsibilities compose. [7].

Sagas established compensation as a way to address partially executed long-lived transactions. Morphogenesis reuses this idea for declared reversible effects. It does not treat compensation as an atomic reversal of arbitrary external actions. Stable operation identities, durable journals, and compare-and-swap are engineering mechanisms used in the composition, not inventions claimed here. [8].

Finally, causal ordering in distributed systems differs from a globally simultaneous observation. Our model records dependencies among observations, decisions, and receipts; it does not reconstruct a universal snapshot or establish that all relevant events have been observed. [9].

## 2.4 Authority, provenance and adaptive design

The confused-deputy problem [14] motivates separating a requester's intent from the authority used to execute it. Vertical Paxos [15] concerns replicated-state reconfiguration; our morphology head consumes storage CAS rather than introducing a new consensus algorithm. W3C PROV [16] supplies a general provenance vocabulary. Our receipts specialize operational dependencies and do not replace provenance semantics or establish that an owner's statement is true.

GPTSwarm [17], AFlow [18] and ADAS [19] optimize agent graphs, workflows or system designs. Such systems can propose changes for an authority-bounded runtime; their optimization results do not establish our cross-owner safety properties. Conversely, protocol safety does not establish the quality of their proposals. The term morphogenesis also has established engineering use in morphogenetic engineering [20]; we claim neither the term nor biological self-organization as a novelty.

Durable Functions has formal semantics for stateful serverless computation and record/replay [21]. Durable progress is therefore a competent baseline, not a distinctive invention here. The proposed contribution is an explicit organizational contract assembled from these established mechanisms, with owner obligations and losing-transition cleanup visible to implementers.

## 3. System model and assumptions

### 3.1 Organization and observation

Let a mission scope be  $s=(\text{tenant}, \text{mission})$ . For analysis, represent a morphology projection at epoch  $e$  as

$$M_e=(s,e,V_e,R_e,T_e,W_e,H_e,P_e).$$

Here  $V_e$  contains references to participants,  $R_e$  to roles and profiles,  $T_e$  to team or topology structure,  $W_e$  to work and continuity records,  $H_e$  to authenticated source heads, and  $P_e$  to policy. This tuple is paper notation, not an additional runtime schema. Each component is a bounded projection or reference to state owned elsewhere.

A source head binds its source identity, revision, digest, scope, authentication evidence, observation time, and expiry. Policy determines the sources required for a particular operator. A missing or stale required source prevents admission. Failure to discover an eligible candidate means absence from this bounded view, not absence from the entire network.

The narrow authoritative morphology head is

$$h_e=(s,e,\text{parentDigest},\text{snapshotDigest},\text{receiptDigest},\text{revision}).$$

It records an accepted organizational successor. It is not the membership registry, the work scheduler, or an authorization database. Subsystem epochs may advance independently; use-time owner checks remain necessary after observation and decision.

### 3.2 Proposals, decisions, and effects

A proposal  $p$  binds the current morphology epoch and digest, a target projection, an operator plan, policy, budget envelope, expiry, and process definition. Its operator plan is a dependency graph  $G_p=(N_p,E_p)$ . Each effectful node names its owning subsystem, stable operation identity, prerequisites, expected receipt, and any declared compensation.

A decision  $d$  approves the exact candidate under a configured route: authorized agent, authorized person, local policy, collective decision, or a composite. Agent membership or proposal authorship alone supplies no decision mandate. Human review is selected by policy rather than required by definition.

We distinguish transition admission from effect admission:

$$\text{Admit}(p,d,h,t)=\text{BindingMatch}(p,d,h)\wedge \text{CurrentRequiredSources}(p,t) \\ \wedge \text{ValidMandate}(d,t)\wedge \text{PolicyAllows}(p)\wedge \text{BudgetReserved}(p,t).$$
$$\text{MayApply}(n,t)=\text{AdmitStillApplicable}(n,t)\wedge \text{OwnerAuthorizes}(n,t) \\ \wedge \text{DependenciesReceipted}(n).$$

Here `BindingMatch` requires equality of the approved proposal, target, scope, predecessor epoch, policy and budget digests. `CurrentRequiredSources` requires every policy-required source to be authenticated, scoped to the mission and unexpired. `ValidMandate` requires an authenticated authorized issuer, the approved actor and route, and an unexpired decision. `PolicyAllows` checks the operator catalog and population/resource limits. `BudgetReserved` requires a live reservation for this proposal and predecessor. `AdmitStillApplicable` rechecks the relevant decision/fence at the admitting boundary. `OwnerAuthorizes` checks the owner's current subject, scope, permitted effect, lease and resource state. `DependenciesReceipted` requires a validated receipt for every prerequisite of the operation. No predicate means that one service can authorize another service's effects.

The experiments use one explicitly controlled clock domain. A logical millisecond value  $t$  maps to UTC origin +  $t$ ; mandate timestamps and controller times are comparable only under that mapping. Real deployments must use an owner-trusted clock, reject regressed local time, and declare their maximum skew or fail closed near an expiry boundary. These experiments do not evaluate skew, wall-clock correction or cross-host leases. An expired approval cannot authorize a new effect; reconciliation of an already applied effect recovers its original evidence rather than granting new authority.

These are specification predicates. Their components are enforced across different boundaries, not by an atomic global check. Correct composition requires each owner to validate the relevant epoch, scope, mandate, and fence when admitting its own effect. A cached positive decision cannot eliminate that requirement.

### 3.3 Failure and trust model

The model allows controller crashes, retries, delayed or lost responses, concurrent proposals, stale observations, and temporary disconnection from required services. An effect may have completed even when its caller received no response. A requester may submit malformed or substituted plans or attempt to reuse expired decisions.

Conditional properties require the following deployment assumptions:

1. Owners authenticate requests and enforce their own scope, epoch, authorization, and resource checks. Protected effects cannot bypass these boundaries.
2. A stable operation identity refers to the same immutable operation. Owners either deduplicate and reconcile that identity reliably or report uncertainty without authorizing blind replay.
3. Durable stores provide the required conditional-write semantics. History rollback is detected using a separately trusted witness where that deployment profile requires one.
4. Digest bindings use canonical serialization and collision-resistant hashing. A digest alone provides neither authentication nor evidence truthfulness.
5. Time and expiry validation follow a documented deployment contract. Progress requires eventual availability of the owners needed for the transition; safety may require indefinite blocking.

Compromised authority owners, stolen signing keys, arbitrary provider dishonesty, unmediated tool use, and undetectable storage rollback are outside these conditional guarantees. The protocol does not establish semantic correctness of an LLM output or prove that a declared capability is useful.

## 4. Transition protocol

Figure 1 separates the mission context, organizational transition, and effect authority. The arrows denote data and control dependencies, not a global transaction.

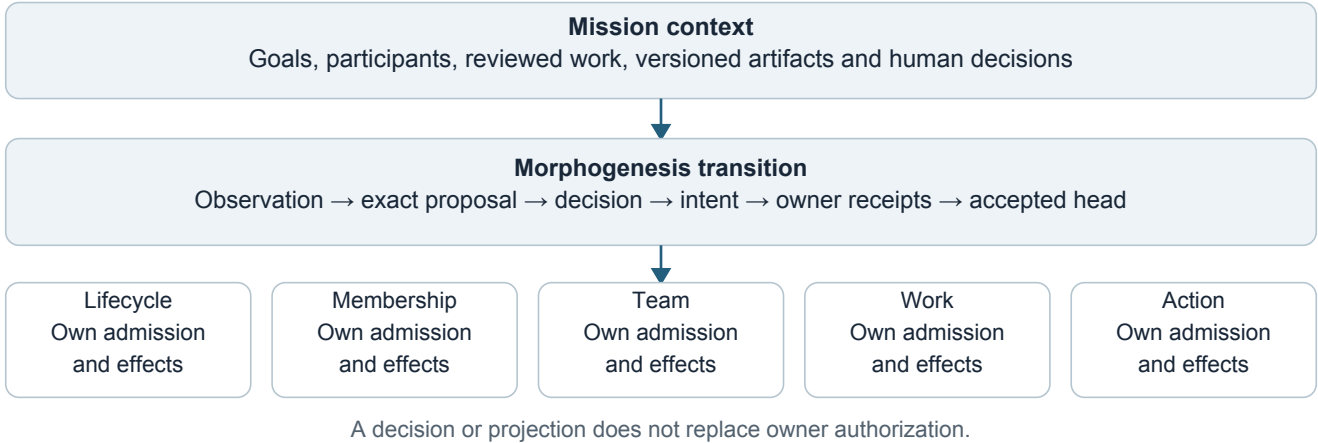


Figure 1. The mission context retains collaboration records. Morphogenesis binds the proposed change and its evidence. Lifecycle, Membership, Team, Work and Action retain their own admission boundaries. An artifact projection grants no operational authority.

### 4.1 Observe, propose, and authorize

The controller validates required source heads and records a bounded need. It selects an admissible target and compiles the operator plan. In the catalog lifecycle profile, the process definition comes from a closed catalog; a proposal cannot install arbitrary workflow code. Advanced operator compilation still uses a closed vocabulary and existing owners.

The proposal binds its source epoch and exact decision subject. A workflow gate may select an approved branch, but branch selection is not sufficient authorization. The controller resolves and validates the decision before preparing effects. The advanced governed entry point additionally binds the compiled plan to a separately issued execution authorization and fence.

Budget reservation accounts for concurrent proposals before resource materialization. It is distinct from observed workflow usage and from budgets enforced by downstream owners. Reservation failure rejects or delays the transition; a reserved global envelope cannot widen an individual work or action grant.

### 4.2 Prepare, enact, and reconcile

Before each external operation, the controller persists immutable intent and its stable identity. On restart, it asks the owner to reconcile any prepared operation whose result is unknown. A confirmed result can supply a validated receipt. A definitely absent effect may be attempted only when the owner contract makes that determination safe against delayed execution. An indeterminate result blocks advancement.

The protocol therefore does not equate a timeout with a failed effect. Nor does an application-side journal itself ensure exactly-once provider behavior. Those properties depend on the effect owner's implementation and the external system it controls.

An abstract execution rule is:

```

for each ready effect in the approved plan:
    persist immutable intent under its stable operation identity
    validate current owner admission requirements
    resolve any previously prepared attempt with the owner
    if its outcome remains indeterminate: stop advancement
    if safely unapplied: request the effect under the same identity
    validate and persist the owner's result receipt
attempt the morphology-head update against the bound predecessor

```

This is explanatory pseudocode. It does not replace the distinct V1 and V2 runtime state machines, nor imply that an owner call and a journal write share a transaction.

### 4.3 Commit and recovery boundaries

The morphology store accepts a successor using revision-and-digest compare-and-swap. A competing update invalidates the expected predecessor; it does not authorize silently rebasing an old decision onto the new state.

Owner effects and morphology commitment are separate events. In the inspected V1 implementation, successor Team activation precedes morphology commitment. Consequently, a controller may recover with an activated Team and an unadvanced morphology head. This intermediate condition must be reconciled from owner receipts. A single winning morphology head is not sufficient evidence that all losing proposals had no material effects.

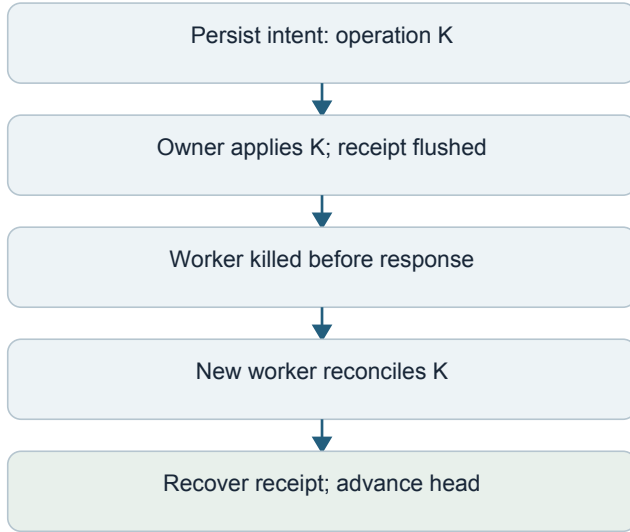
Declared reversible effects may be compensated before commitment. The advanced profile journals compensation separately and processes applied steps in reverse order. Irreversible effects and unresolved outcomes require explicit handling; they cannot be erased by restoring an old projection. After commitment, recovery is represented as a successor transition rather than an atomic rollback of history.

### 4.4 A losing proposal can leave an effect

Consider two proposals A and B bound to morphology epoch *e*. Each obtains an owner receipt for Team activation before either updates the morphology head. A commits first. B's update is rejected as stale, but B's owner receipt still exists. Section 7 first isolates this boundary with synthetic effects, then exercises two approved proposals through the reference Team adapter, Work reducers and PostgreSQL stores.

The correct observable is therefore a pair: accepted successors and material effects from losing proposals. Reporting only one accepted successor hides the second quantity. A deployment must specify whether pre-commit Team activation can admit useful work, which owner fences constrain it, and how losing effects are reconciled or compensated. This paper does not assert that V1 automatically compensates every such effect. The integrated experiment uses two explicit local Work grants; it does not bypass those owners. In the frozen v0.3 experiment, cleanup cancels the losing Team and releases its Work contract, but that older V1 execution remains in `committing_morphology`. This distinction separates cleanup of effects from termination of the coordinating state machine.

### A. Lost acknowledgement



### B. Competing successors

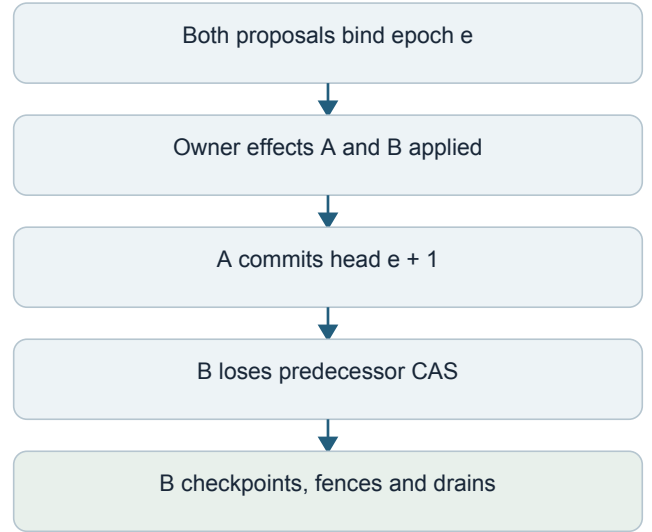


Figure 2. Left: durable intent and an owner receipt survive worker termination; reconciliation recovers the same operation. Right: two owner effects may precede a single accepted head. Compensation or successor recovery is a separate obligation.

## 4.5 Terminal supersession

The updated reference runtime distinguishes an accepted transition from a proposal displaced by another successor. Resolution requires an authoritative head at exactly the proposal's expected epoch plus one, with another accepted proposal and matching scope. The execution must have an activated Team and no committed activation receipt. An arbitrary later head or a timeout is insufficient proof; such cases remain blocked until historical evidence can establish the outcome.

The runtime durably binds the observed winning head, then checkpoints retained work, fences future authority, drains admitted work, detaches or retires the participant, and releases the reservation. Only then does it issue a separate supersession receipt and enter the terminal state superseded. An unknown owner result retains the prepared operation identity for reconciliation. Neither cleanup nor the terminal receipt rewrites the winner or claims that the losing proposal activated the morphology. Legacy normal records retain their field set and digests; the new states are opt-in.

## 4.6 Work continuity and retirement

Detachment and retirement require a continuity checkpoint and fencing of work/action authority before removing or terminating the resource. The checkpoint identifies useful work and its provenance; successor work allocation remains owned by the work subsystem. Material availability, membership, team participation, and permission to act are distinct states.

For the specialist example, the transition records why the agent was added, which profile and decision authorized the plan, which work it received, and which receipts established its later removal. If a retirement acknowledgement is lost, reconciliation uses the original operation identity. Causal continuity means these dependencies remain reconstructible; it does not mean that every mission succeeds or that every generated artifact remains semantically useful.

## 4.7 Three meanings of continuity

For an accepted transition, define evidence continuity as the ability to resolve a typed dependency path from predecessor and observation through proposal, decision, intent, owner receipts, and successor. Every edge must have matching scope and immutable subject bindings. Hash agreement alone does not establish source authenticity or retention.

Work continuity concerns artifacts: for a nonempty set of predecessor artifacts required by successor tasks, measure the fraction still addressable at the required version with valid dependencies. An empty required set is

reported as not applicable, not perfect continuity. Semantic validity needs a task oracle; existence of an artifact is insufficient.

Mission continuity concerns purpose and constraints across multiple transitions. It is outside the core evaluation. Here, causal continuity denotes execution provenance and dependencies, not a causal estimate of performance improvement.

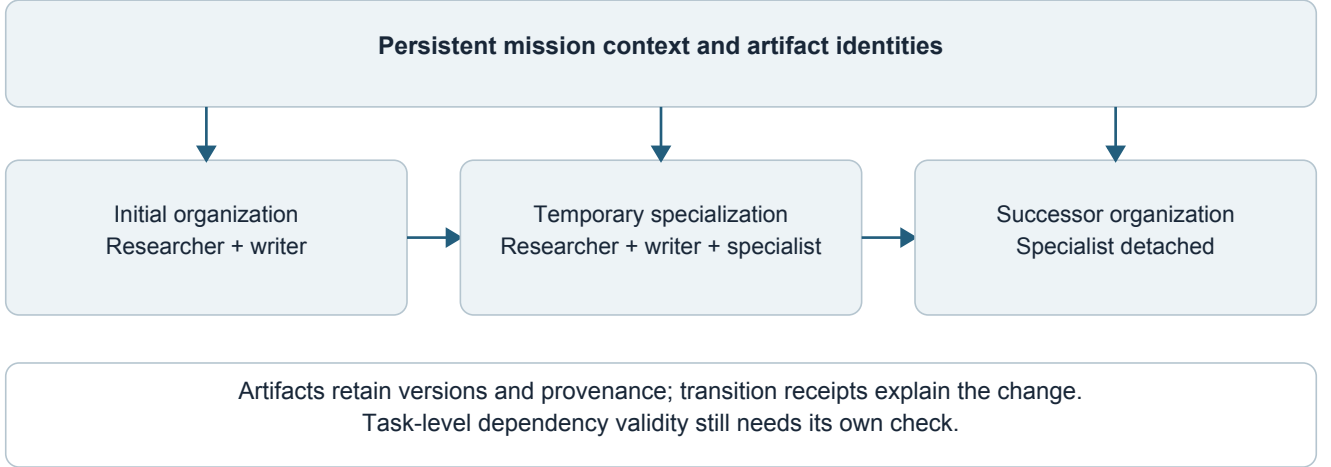


Figure 3. The mission context and artifact identities can persist while participants change. An accepted receipt explains a transition; artifact dependency checks determine whether prior work remains useful. Neither property alone proves mission fidelity.

## 5. Conditional properties and proof obligations

The following arguments apply to the abstract model under Section 3's assumptions. They are proof sketches and implementation obligations, not machine-checked theorems about the entire repository.

**P1 — Owner-bounded authority.** Every admitted protected effect has a current grant from its owner. Assuming complete mediation and correct owner validation, induction over effect admissions establishes this property: coordination records create no alternative admission path. An implementation that invokes an unprotected provider bypasses the premise. This property does not imply that an authorized policy is ethically appropriate or that authority can never be expanded through a legitimate owner decision.

**P2 — One accepted successor per predecessor revision.** If all writers use the same authoritative, linearizable conditional store and the expected revision/digest pair, at most one distinct successor can consume that pair. The first successful write changes the comparison value; a competing write cannot succeed against it. This is a local head property. Restored stores, disconnected authority replicas, and material side effects require additional mechanisms and are not covered by CAS alone.

**P3 — No blind replay of unresolved effects.** If every effect is preceded by durable intent, recovery inspects that intent, and indeterminate owner results block execution, controller recovery cannot intentionally resubmit the unresolved operation as a fresh operation. Preventing duplicate material effects additionally requires correct owner deduplication and reconciliation. A delayed original request racing with an incorrect “absent” answer is a counterexample when those assumptions fail.

**P4 — Terminal removal follows fencing.** If the terminal boundary requires validated checkpoint and fence receipts and the owners enforce those fences at action admission, resource removal follows the declared continuity and revocation steps. Already admitted in-flight actions require the owner's draining contract; fence issuance does not retroactively undo them.

**P5 — Accepted lineage is evidence-linked.** If the head accepts only validated receipt bindings and predecessor references, each accepted successor has a reconstructible dependency path to its proposal and decision. Hash linkage detects some inconsistencies but does not prove that an external receipt is truthful, complete, or durably retrievable. Evidence retention is a separate operational obligation.

| Property      | Runtime obligation                                        | External premise and counterexample                                         |
|---------------|-----------------------------------------------------------|-----------------------------------------------------------------------------|
| P1: authority | Route protected effects through their owners              | Correct use-time grants; a bypassing tool call defeats mediation            |
| P2: successor | Compare the expected predecessor in the head store        | Linearizable store; effects from a losing proposal remain possible          |
| P3: recovery  | Retain intent and operation identity; stop on uncertainty | Reliable owner reconciliation; a false absent response permits duplication  |
| P4: removal   | Require checkpoint and fence before terminal removal      | Owners enforce fences and draining; already-admitted work can remain active |
| P5: evidence  | Validate predecessor, decision and receipt bindings       | Retain authentic evidence; a valid digest cannot recover a deleted object   |

The table allocates obligations rather than assuming that composition automatically establishes them. In particular, a synthetic owner rejection demonstrates that a denial propagates through the coordinator, not that an independently deployed owner correctly implements authorization.

Liveness is intentionally weaker than safety. An unavailable owner, expired decision, lost budget reservation, or indeterminate effect can block progress. A meaningful evaluation must count those blocked missions and their recovery cost instead of treating all refusals as successful adaptation.

## 6. Reference implementation

AgentPlat implements Morphogenesis as an opt-in subpath of AgentPlat Collective Runtime. Domain contracts represent observations, proposals, decisions, execution records, and receipts. Governed Durable Workflows owns process execution. The Collective Host PostgreSQL adapter owns durable morphology storage. Agent Rooms and Agent Mesh expose projections without acquiring execution authority. The package architecture preserves existing lifecycle, Membership, Team, Work, Trust, and Action boundaries. Implementation mappings and architecture decisions are indexed in the artifact.

The paper's core scope corresponds to the V1 specification, with compiled operators and compensation from V2. Selected source inspection confirms explicit plan/decision matching and authorization expiry checks in the governed V2 entry point, prepared-operation reconciliation in the V1 execution runtime, and an indeterminate compensation state in V2. These observations establish source structure, not deployment-wide enforcement.

The implementation uses content-minimized control records: identifiers, enums, counters, and digests rather than raw prompts, model outputs, or private keys. Such records can still expose metadata and correlations. Content minimization is not a general privacy proof.

### 6.1 Agent Room integration

An Agent Room supplies persistent goals, participants, versioned artifacts, approvals and provenance [12]. AgentPlat projects transition proposals and outcomes as Room artifacts. Its participation adapter adds a participant with authority level zero and no permissions; it does not mint Mesh membership or Work authority. The authorized-person decision adapter binds a Room approval to an exact candidate and mandate. Approval of an unrelated artifact cannot stand in for that decision.

The implementation's `roomId` may be null. Rooms are therefore a concrete integration context, not a requirement of every protocol deployment. In the running example, the Room retains the proposal's work while a specialist joins and later leaves the active organization. The local pilot checks artifact addressability across a transition; semantic dependency validity, participant lifecycle synchronization and durable Room recovery require additional integration evaluation.

## 6.2 Version and capability boundaries

The v0.2-v0.4 evaluations retain their frozen source, compiled-module and lockfile bindings. The v0.5 extension uses a separately recorded implementation with explicit supersession. The artifact index provides exact revisions and reproduction commands. Later strategy synthesis and constitutional-evolution APIs are outside this paper's evaluated scope.

## 7. Local evidence: boundary and integration pilots

### 7.1 Boundary pilot method

Eleven prescribed cases exercise the V1 runtime with a synthetic Team owner, the real morphology-head implementation and a flushed single-writer JSONL test adapter. For a lost acknowledgement, the worker receives SIGKILL after the owner receipt is flushed; a distinct process then reopens state. The V1 runtime is compared with a fresh-identity retry ablation and a small write-ahead/stable-ID controller. The comparator covers this one boundary, not a complete saga framework. Counts are compared, not performance.

The design preceded execution; each schedule has one retained run. Two development attempts failed on a Room artifact/version lookup mistake and remain in the artifact. The v0.2 pilot/ bundle retains raw effects, journals, process identities, receipts and source hashes. Its separate verifier recounts effects and validates bindings. These are local diagnostic results, not independent replication.

### 7.2 Observations

| Boundary and condition                         | Observed outcome                                           | Interpretation                                                 |
|------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------|
| Nominal: all three conditions                  | One Team effect each                                       | Common nominal behavior at this boundary                       |
| Lost response: V1 runtime                      | One effect after worker replacement                        | Stable-ID reconciliation recovers the receipt                  |
| Lost response: durable-pattern comparator      | One effect after worker replacement                        | Recovery benefit is shared with an established durable pattern |
| Lost response: fresh-identity ablation         | Two effects on the same semantic Team target               | A new identity escapes per-operation deduplication             |
| Owner unavailable during recovery              | Blocked before head advancement; resumed after restoration | Unknown outcome does not become fresh execution                |
| Current owner denies admission                 | No Team effect; owner rejection retained                   | Earlier approval does not override the synthetic owner         |
| Two effects followed by competing head updates | One accepted successor, one losing-proposal effect         | Head consistency is weaker than transition atomicity           |
| Detachment attempted before fence              | Rejected; normal completion after fencing                  | V1 enforces the tested phase prerequisite                      |
| Room artifact/receipt projection               | Earlier work version retained; receipt retrievable         | Addressability in the in-memory Room repository                |

The competing-head probe applies two synthetic owner effects and then submits competing commits to the real head runtime. It does not execute two complete independently approved transitions and does not test automatic cleanup. The owner-denial fixture returns a fixed rejection after organizational approval; it does not exercise a real mandate-expiry race. Similarly, the Room case does not evaluate semantic artifact quality. The pilot intentionally reports these narrower boundaries instead of turning them into broader system guarantees.

### 7.3 Interpretation

Stable-ID recovery is shared with an established durable pattern; the pilot does not establish unique recovery superiority. Its losing-effect trace shows why an accepted head alone is insufficient. The organizational contribution is the explicit composition of decisions, owners, effects and accepted lineage.

### 7.4 PostgreSQL integration: competing transitions, expiry and drain

The second pilot replaces the synthetic Team owner with the reference Team Formation runtime and its Morphogenesis adapter. It uses the production PostgreSQL Morphogenesis head, execution and budget adapters, the Collective Work reducers, and the PostgreSQL authority/execution repositories. Team persistence uses an experiment-specific PostgreSQL CAS adapter. Discovery, membership observations and capability attestations remain fixtures. This is not a complete distributed Mesh deployment.

Two exact proposals bind the same predecessor, with distinct target digests and locally signed Ed25519 approvals from different test reviewers. The approved target explicitly binds the Team-position mapping used by the adapter. Work contracts are registered through the real owner reducer after a current mandate check. Each Team activation produces a small local work artifact before morphology commitment. These are instrumented application work products, not external ActionGateway effects.

| Observation                    | Recorded result                                                                                               | Scope of conclusion                                                                   |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Competing approved proposals   | Two active Teams and two registered Work contracts before head advancement; one accepted morphology successor | Owner-valid effects can precede the losing morphology CAS                             |
| Explicit losing-owner cleanup  | Work released and Team cancelled; V1 execution still committing_morphology                                    | Owner cleanup does not imply terminal coordinator state                               |
| Work mandate expiry            | At owner time origin + 380 ms, decision validity ends at +470 ms and mandate validity at +350 ms              | Real authority reducer rejects Work admission with mandate_expired                    |
| Detachment with admitted work  | Fence rejects new work; detachment stays pending until the admitted application job finishes                  | Core Work revocation plus the application's explicit drain contract                   |
| Recovery of pending detach     | Original operation ID reconciled; winner reaches completed and releases its budget                            | Tested lifecycle completes without treating fencing as cancellation of in-flight work |
| New PostgreSQL connection pool | Head and Work digests agree after reopening                                                                   | Local persistent state; no host-loss claim                                            |

For the expiry case, the Team proposal remains awaiting\_member\_contracts, no Work contract is registered and the morphology head stays at epoch 1. The fixture therefore distinguishes a valid organizational decision from an expired effect-owner mandate. For draining, the application deliberately holds a previously admitted promise open; revoking its Work contract blocks new admissions but does not erase that promise. The detachment adapter enforces waiting. That adapter obligation is not an unconditional guarantee supplied by Morphogenesis alone.

The competing-transition result motivates a precise operational requirement: supervise both residual owner effects and nonterminal execution records. The experiment's cleanup is explicit application orchestration using existing owner APIs; it is not automatic V1 compensation. The original work artifacts remain in the evidence after cleanup. A future terminal cancellation/supersession protocol must define how a losing execution becomes safely non-retryable while retaining those effects and their resolutions.

The integration/ bundle retains complete proposal/decision bindings, public approval keys and signatures, owner states before and after cleanup, execution records, artifact rows, expiry/drain traces and source/module

hashes. The local keys establish fixture integrity, not real-world reviewer identity. The rollback witness is process-local and gives no independent protection against host loss or database rollback. A separate verifier validates hashes, approval signatures, runtime records and causal event ordering, with negative tests for rehashed tampering. The first development attempt failed because the late Team bid's completion window was invalid; that trace is retained. Draining was added as an explicitly documented pilot extension before its runs.

## **8. Four-condition controlled mission evaluation**

### **8.1 Design and fairness**

The final evaluation asks how transition controls affect correct mission output, reservation use, blocked progress and persistence overhead. It uses eight-phase integer-array missions with four demand shapes: stable general work, a specialist burst, shrinking demand and a false specialization forecast. Workers compute sums or exact population variances. A late input revision supersedes an earlier artifact, and the final report must reference the correct versions and values.

The four conditions share inputs, deterministic workers, role catalog, owner guard and immutable artifact store. The fixed organization starts with the minimal capability set sufficient for the entire actual workload, giving it advance knowledge rather than making it incapable by construction. Minimal adaptation retries a lost activation using a fresh identity. The durable baseline uses AgentPlat's actual workflow runner over its PostgreSQL store, exact process inputs, bounded task retries and stable owner-operation reconciliation. Morphogenesis uses its V1 execution runtime and PostgreSQL execution/head adapters. All adaptive conditions receive the same explicit need/forecast schedule and defer retirement while required work is outstanding. The schedule is supplied, not learned.

The mission owner is instrumented: discovery, capability identity, decision issuance and the Morphogenesis reservation binding are fixtures. Application data tasks use the common artifact owner rather than the Agent Room API. This study complements the separate Room projection and Team/Work integration experiments; it does not combine them into a fully deployed end-to-end platform.

Each condition has a ceiling of 16 role-phase units and 512 client SQL calls, including governance persistence; two SQL calls are reserved for rollback. A generalist counts in every phase, and materialized duplicate reservations consume the same role budget. These are separate resource dimensions, not dollar costs. Timing and SQL counts cover per-mission owner/controller execution, including initialization and attempted final cleanup; shared schema/catalog setup and evidence export are excluded. SQL counts include transaction-control statements and are not physical storage I/O measurements.

Three schedules inject no fault, a lost activation acknowledgement, or a lost activation acknowledgement followed by unavailable reconciliation for two phases. The fixed organization has no later activation opportunity, an exposure difference reported explicitly. Both durable conditions retain uncertainty and allow unrelated general work to continue while the transition is blocked.

### **8.2 Pilot, freeze and independent scoring**

Two initial 96-run development passes checked instrumentation and added a common query ceiling and per-query trace. A final 96-run engineering pilot used seeds 9001 and 9002. All three passes are retained separately and excluded from the final dataset; the role ceiling was not tuned to favor a condition. Before final execution, a local registration froze source hashes, inputs, budgets, exclusions and analysis. Seeds 101-132 generate 32 held-out input arrays per shape/fault cell, paired across the four conditions: 1,536 runs. This is finite-grid coverage, not a power calculation or a public preregistration.

A separate Python oracle imports no JavaScript worker/controller code. It computes exact answers with rational arithmetic, verifies every artifact digest, checks current input versions and final-report dependencies, and reconstructs role occupancy from materialization/retirement events. Negative tests reject wrong arithmetic and obsolete dependencies even after hashes are recomputed. A separate JavaScript verifier checks matched inputs, runtime/workflow records and every prescribed fault opportunity. Unexpected errors invalidate an attempt; denials, blocked phases and budget exhaustion remain measured outcomes. No case was excluded from the held-out run.

### 8.3 Results

| Condition          | Correct reports | Duplicate reservations | Mean role-phase units | Median SQL calls |
|--------------------|-----------------|------------------------|-----------------------|------------------|
| Fixed              | 384 / 384       | 0                      | 12.0                  | 22.0             |
| Minimal adaptation | 320 / 384       | 192                    | 12.0                  | 25.0             |
| Durable workflow   | 384 / 384       | 0                      | 9.5                   | 96.5             |
| Morphogenesis      | 384 / 384       | 0                      | 9.5                   | 69.0             |

Each adaptive condition encountered 192 fault-exposed missions; fixed encountered none. Fresh-identity retries produced 192 duplicate role reservations in the minimal condition. In the shrinking-demand cells, the untracked reservation continued consuming capacity: 64 runs exhausted the role budget and missed one required task and the final report. Both durable conditions accumulated 192 blocked phases while reconciliation was unavailable, then completed all reports after restoration. No query ceiling was reached.

The two durable conditions have the following role-phase consumption in every retained seed within each cell:

| Demand shape   | No fault | Lost acknowledgement | Lost acknowledgement + reconciliation outage |
|----------------|----------|----------------------|----------------------------------------------|
| Stable         | 8        | 8                    | 8                                            |
| Burst          | 10       | 10                   | 11                                           |
| Shrinking      | 10       | 10                   | 11                                           |
| False forecast | 9        | 9                    | 10                                           |

The no-fault mean is 9.25. Across the twelve demand/fault cells it is 9.5 because delayed reconciliation adds one reservation phase in three cells. The full 48-cell table, including all four controllers, is in the artifact. Thirty-two seeds vary arithmetic inputs; the invariant protocol outcomes within cells are not independent reliability observations. The matrix has 48 design cells, not 1,536 independent tests of general safety.

Stratification changes the interpretation of the aggregate. With no fault, all conditions use 8 role-phase units for stable work. For burst or shrinking demand, fixed uses 16 and each adaptive condition uses 10. A false forecast reverses the benefit: fixed uses 8 and each adaptive condition uses 9. The reservation advantage therefore belongs to the chosen adaptation policy under particular demand, not uniquely to Morphogenesis.

All accepted artifacts remained addressable, including those in failed minimal runs. Addressability alone did not ensure a complete report with valid dependencies: the oracle accepted 384 final reports in each of the three successful conditions and 320 in the ablation. This operationalizes the distinction between retaining evidence and completing useful work.

Median local elapsed time was approximately 7.0 ms for fixed, 7.6 ms for minimal, 36.9 ms for the workflow baseline and 45.6 ms for Morphogenesis. The ratio of these medians is approximately 1.24, but this is not a general overhead estimate. SQL count measures client statements, whereas elapsed time also includes validation, hashing, serialization, scheduling and database latency. The historical run did not attribute time among these components, so its data cannot establish which explains the difference. These single-host micro-workload timings are diagnostic. Fewer SQL statements did not imply lower elapsed time here; the study does not identify a general performance ranking or a monetary break-even point.

## 8.4 Interpretation and reproduction

The competent durable baseline matches Morphogenesis on correct task output, duplicate prevention and role reservations. The observed benefit over minimal coordination does not establish unique superiority of the proposed protocol. Its distinct research object remains the explicit organizational/authority/evidence contract. The experiments reveal both what the composed implementation does and what it leaves to adapters and operational recovery.

The companion artifact provides the locally frozen registration, all 1,536 raw cases, separate pilot data, paired descriptive differences, per-cell summaries and source-bound verifiers. The mission design and scripts reproduce the study using local PostgreSQL without paid inference. The workload has a small role catalog, deterministic task semantics, fixture authorities and prescribed faults. LLM proposal quality, multi-tenant deployment, independent hosts and typical production economics require separate studies.

## 9. Governance and supersession extension

### 9.1 Controlled authority matrix

We run nine prescribed cells against the four controllers: nominal admission, expired decision, mandate expiry after approval, plan substitution, cross-scope escalation, self-issued authority, owner denial, lost acknowledgement, and lost acknowledgement with reconciliation unavailable. All controllers use the same signed admission envelope, configured issuer trust anchor, PostgreSQL owner and use-time checks. The durable comparator is the real workflow runner with PostgreSQL persistence. The Morphogenesis condition uses the reference V1 execution/head runtime. The owner envelope is experiment-specific adapter code; it is not a claim that the core runtime supplies identity or mandate enforcement automatically.

| Condition        | Invalid-authority cells rejected | Materialized effects in three positive/recovery cells | Valid exported admission chains | Reconstructible terminal controller chains |
|------------------|----------------------------------|-------------------------------------------------------|---------------------------------|--------------------------------------------|
| Fixed            | 6 / 6                            | 3                                                     | 3 / 3                           | Not applicable                             |
| Minimal          | 6 / 6                            | 5                                                     | 5 / 5                           | Not evaluated                              |
| Durable workflow | 6 / 6                            | 3                                                     | 3 / 3                           | 3 / 3                                      |
| Morphogenesis    | 6 / 6                            | 3                                                     | 3 / 3                           | 3 / 3                                      |

No invalid-authority cell materializes an effect in any condition. Both adaptive durable conditions retain one effect per positive/recovery cell; minimal fresh-identity retry produces two extra effects. Fixed materializes at initialization and has no later fault exposure. These are 36 prescribed cells, not estimated attack probabilities. No LLM generates the proposals. The experiment isolates enforcement against deterministic adversarial requests rather than open-ended agent intelligence.

An independently implemented verifier checks the configured issuer signature, exact proposal/scope/subject, time validity, exported owner admission and effect identity. It recomputes Morphogenesis record and receipt digests and checks event links. The workflow comparison checks the matching run, successful activation/retirement stages and fenced inactive role. Nine verifier tests include a valid control and eight mutations: missing chain, scope substitution, changed proposal, expired time, forged signature, missing fence, changed terminal phase and missing admission. All mutations are rejected.

These metrics operationalize admission-chain and terminal-controller reconstruction. They are not a complete comparative measurement of P5 under contention: the workflow adapter does not export the same morphology-head schema, and the competing-successor experiment below exercises Morphogenesis alone. That boundary prevents interpreting different record formats as a safety advantage. A full cross-controller contention/provenance benchmark remains open.

## 9.2 Persistent supersession and worked trace

The updated executable assembly connects real PostgreSQL Room approval, Team Formation, Work authority, execution and head stores. Catalog/membership observations and scripted reviewers remain fixtures; the rollback witness remains process-local. Two proposals activate their Teams before competing for one head. Exactly one advances the head. The other is fenced, detached, budget-released and recorded as superseded. A fence acknowledgement is lost after the Work mutation but before the adapter receipt; reopening the runtime reconciles the original operation from durable state. A separate database connection reads the resulting head and Work state.

| Trace step              | Retained observation                                                                         | Consequence                                                          |
|-------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Approval and activation | Two distinct approved proposals; two Teams and Work contracts                                | Both owners admitted effects before head selection                   |
| Head competition        | Epoch 1 advances once to epoch 2                                                             | One proposal wins; the other's effect is still real                  |
| Loser resolution        | Checkpoint, original fence identity recovered, Team cancelled, Work revoked, budget released | Separate terminal supersession receipt; no normal activation receipt |
| Winner retirement       | New work rejected after fence; detach remains pending while one admitted job runs            | Completion requires that job to finish                               |
| Expiry control          | Organizational decision valid at +380 ms; owner mandate expired at +350 ms                   | No Work contract admitted; head remains unchanged                    |

The artifact includes the actual execution records and approvals behind this trace. A separate verifier reconstructs both terminal receipt chains and their owner postconditions. It does not infer production identity trust from fixture signatures. The result closes the specific nonterminal-loser counterexample reported by the frozen older implementation, without rewriting its evidence.

## 9.3 Bounded model checking

A TLA+ abstraction has two proposals, one predecessor, one stable effect identity and at most one admitted job per proposal. TLC 1.7.4 explores 260 distinct states (761 generated, depth 17) and finds no violation of successor uniqueness, at-most-one effect, fenced/drainage detachment or terminal effect presence. Under explicit weak fairness of proposal steps, reconciliation and job completion, both transitions eventually close.

Three deliberately weakened configurations produce counterexamples: fresh-identity retry violates the stable-effect invariant; detachment before draining violates safe termination; omission of loser resolution violates eventual closure. Complete model, configurations, logs and counterexample traces are retained. Owner atomicity, durable identities and fair recovery are assumptions. The model abstracts authorization cryptography, multiple epochs and clock skew; it is not a refinement proof of the TypeScript implementation.

## 10. Limitations and future work

The literature comparison is selective and does not prove priority over all organizational middleware, runtime verification or distributed reconfiguration. The predicates now have explicit obligations, but a formal refinement mapping to the full runtime remains absent. The bounded model and executable supersession path address the identified losing-transition gap under their stated assumptions; neither is an end-to-end formal proof.

The trusted computing base is substantial: identity issuers, policy configuration, effect owners, storage semantics, expiry handling, and deployment wiring. A correct library can be composed incorrectly. Resource reservation also cannot bound activity that bypasses the registered resource authorities. Local multi-process tests do not model independent physical failure domains or long-lived production traffic.

Causal evidence does not establish semantic mission fidelity. A formally authorized organizational change can be unhelpful, expensive, or based on mistaken capability evidence. The finite-grid task oracle addresses exact

arithmetic and dependency correctness; it does not assess open-ended LLM output quality or learned adaptation policies. Human or quorum decisions introduce additional costs and failure modes, including correlated reviewers and inadequate mandates.

Later AgentPlat versions explore strategy adaptation, synthesized profiles, agent genesis, whole-collective evolution, and constitutional continuity. These extensions raise a longitudinal question: can individually admissible changes accumulate into mission drift or excessive authority concentration? That question motivates future work but is not answered by the V1/V2 evidence here. Finite model exploration, when used, must report its bounds and incomplete outcomes rather than claiming unrestricted proof.

## 11. Conclusion

Governed agent morphogenesis treats organizational change as a coordinated transition among independently authoritative subsystems. Its central separation is between proposing a new organization, authorizing a bounded plan, and admitting each material effect. Durable intent, owner receipts, conditional successor selection, and explicit recovery make that separation inspectable across failures. AgentPlat supplies a reference implementation and separately versioned local evaluations. The pilots show shared stable-ID recovery behavior, owner-mandate expiry enforcement, and the distinction between one accepted successor, residual losing effects and nonterminal coordinator state. The new supersession path closes the losing execution only after owner cleanup; the controlled authority matrix finds equal admission safety with an equivalently guarded workflow baseline. The controlled mission study identifies reservation tradeoffs and matches a competent durable baseline on task outcomes. General organizational intelligence and deployment reliability remain outside these results.

## Artifact availability

The public AgentPlat repository contains the manuscript sources, reproducible experiments and raw evidence [1]. The v0.5 artifact index separates the frozen v0.4 mission matrix from the new governance cells, persistent supersession trace and bounded model. Source and compiled-module digests identify executed code; the index carries exact version references and commands. All new experiments run locally without model credentials or paid inference.

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